

A Minor Project Report On

**Automated Two-Wheeler Violation Detection
and E-Challan System**

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Abstract

The Urban traffic management is increasingly challenged by rising vehicle density, limited enforcement resources, and growing road safety violations. This project proposes a Smart Rider Monitoring & Violation Detection System that leverages real-time computer vision and deep learning techniques to automatically identify traffic rule violations involving two-wheelers. The system focuses on critical infractions such as helmet non-compliance, triple riding detection, and vehicle number plate recognition for automated enforcement.

The proposed architecture processes live CCTV or RTSP video streams through a modular AI pipeline consisting of vehicle detection, rider detection, helmet classification, and optical character recognition (OCR) for number plate extraction. YOLOv8-based object detection models are utilized for accurate and efficient detection tasks, while EasyOCR/PaddleOCR is integrated for robust text extraction from number plates. A rule-based violation engine evaluates detected events and generates evidence including images, timestamps, and location metadata.

The system is designed for real-time performance using optimized inference pipelines, GPU acceleration, and optional edge deployment on devices such as NVIDIA Jetson modules. A backend system built with FastAPI and a dashboard interface enables live monitoring, alert generation, and violation analytics.

The proposed solution aims to improve traffic law enforcement efficiency, reduce human dependency, and enhance road safety through scalable and automated AI-driven surveillance.

Keywords: YOLOv8, Computer Vision, Traffic Surveillance, Helmet Violation Detection, Triple Riding Detection, ANPR, OCR, RTSP, Deep Learning, Smart Traffic Management.

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1 Introduction

Road safety remains a major concern in modern transportation systems, with traffic violations contributing significantly to accidents, injuries, and loss of life. Among two-wheeler riders, violations such as riding without helmets, carrying more than the permitted number of passengers, and failure to comply with traffic regulations are common challenges faced by traffic authorities. Traditional enforcement methods primarily depend on manual monitoring, which can be inefficient, labor-intensive, and difficult to scale in areas with high traffic density.

The Automated Two-Wheeler Violation Detection and E-Challan System is an intelligent traffic monitoring solution designed to automatically identify traffic rule violations involving motorcycles and scooters. By utilizing computer vision and artificial intelligence, the system continuously analyzes video streams to detect violations, identify vehicles, and generate digital evidence for enforcement purposes. The integration of automated number plate recognition and e-challan generation enables a faster, more transparent, and more efficient enforcement process.

The vision of this project is to support the development of smarter and safer transportation systems through technology-driven traffic management. By reducing dependence on manual surveillance and enabling real-time violation detection, the proposed system aims to improve compliance with traffic regulations, enhance road safety, and contribute to the advancement of intelligent transportation infrastructure.

a. Problem Statement

Road traffic violations remain a significant challenge in urban transportation systems worldwide. Among two-wheeler riders, violations such as riding without helmets, carrying more than the permitted number of passengers (triple riding), and non-compliance with traffic regulations contribute substantially to road accidents and fatalities. Traditional traffic monitoring and enforcement methods rely heavily on manual observation by traffic personnel, making the process time-consuming, labor-intensive, and prone to human error.

With the rapid increase in vehicle numbers, continuous monitoring of traffic activities has become increasingly difficult. Existing surveillance systems often record traffic footage without providing automated mechanisms for detecting violations and identifying offenders. Consequently, many violations go unnoticed, reducing the effectiveness of traffic law enforcement and compromising road safety.

Therefore, there is a need for an intelligent and automated system capable of monitoring traffic in real time, detecting violations accurately, extracting vehicle information, and generating evidence for enforcement purposes.

b. Project Objectives

The primary objective of this project is to design and develop a Smart Rider Monitoring & Violation Detection System using computer vision and deep learning techniques for automated traffic surveillance.

The specific objectives are:

- To detect motorcycles and riders from live CCTV or RTSP video streams.
- To identify helmet violations among motorcycle riders.
- To detect triple-riding violations by estimating the number of riders on a motorcycle.
- To recognize vehicle number plates using Optical Character Recognition (OCR).
- To generate violation records with supporting evidence such as images, timestamps, and vehicle information.
- To provide a real-time monitoring dashboard for viewing detected violations and system analytics.
- To improve traffic law enforcement efficiency through automation and AI-based surveillance.

c. Project Scope

The scope of the proposed system includes the development of an intelligent traffic monitoring platform focused on two-wheeler violation detection.

The system will:

- Process live video feeds from CCTV cameras or RTSP streams.
- Detect motorcycles and riders using YOLOv8 object detection models.
- Identify helmet compliance and triple-riding violations.
- Extract vehicle registration numbers using OCR techniques.
- Store detected violations in a centralized database.
- Generate alerts and maintain violation records through a web-based dashboard.
- Support real-time processing using GPU acceleration and optimized inference pipelines.

The project is limited to monitoring two-wheeler traffic violations and does not cover violations such as overspeeding, signal jumping, illegal parking, or lane discipline enforcement.

d. Limitations

Although the proposed system aims to provide accurate and automated traffic violation detection, several limitations exist:

- Detection accuracy may decrease under poor lighting conditions, rain, fog, or low-quality video footage.
- Occlusion caused by other vehicles, pedestrians, or traffic congestion may affect rider and helmet detection performance.
- Number plate recognition accuracy may vary due to motion blur, damaged plates, unusual fonts, or unfavorable camera angles.
- The system's effectiveness depends on camera placement, resolution, and field of view.
- Real-time performance may require dedicated GPU hardware for high-resolution video streams.
- The model may require periodic retraining to maintain accuracy across different traffic environments and vehicle types.

- The current implementation focuses only on two-wheeler violations and does not detect all categories of traffic offenses.

e. Significance of Study

The proposed Smart Rider Monitoring & Violation Detection System contributes to the advancement of intelligent traffic management by integrating computer vision, deep learning, and automated surveillance technologies.

The significance of this study includes:

- Enhancing road safety by promoting compliance with helmet and passenger safety regulations.
- Reducing dependency on manual traffic monitoring and enforcement.
- Increasing the efficiency and consistency of traffic violation detection.
- Providing real-time monitoring capabilities for traffic authorities.
- Supporting data-driven decision-making through violation analytics and reporting.
- Demonstrating the practical application of artificial intelligence in intelligent transportation systems.
- Establishing a scalable framework that can be extended to detect additional traffic violations in future developments.
- Contributing to the adoption of smart-city technologies for modern traffic management and enforcement.

This study demonstrates how AI-powered surveillance systems can improve traffic law enforcement while reducing operational costs and enhancing public safety.

2 Literature Review

Road traffic violations are a major cause of accidents and fatalities worldwide. With the advancement of computer vision and deep learning technologies, automated traffic monitoring systems have gained significant attention for detecting violations such as helmetless riding, triple riding, mobile phone usage while driving, and vehicle identification through Automatic Number Plate Recognition (ANPR).

Recent studies have demonstrated the effectiveness of deep learning-based object detection models for traffic surveillance applications. Among various approaches, the YOLO (You Only Look Once) family of models has emerged as one of the most widely adopted techniques due to its high detection accuracy and real-time processing capability. Chen et al. proposed a deep learning framework for detecting helmet-rule violations among motorcyclists and reported promising results under complex traffic conditions [1]. Similarly, Said et al. enhanced helmet detection performance using an improved PerspectiveNet architecture integrated with EfficientNetV2, achieving high accuracy in motorcycle rider classification tasks [2].

Helmet violation detection has been extensively studied because the use of helmets significantly reduces the risk of fatal injuries in motorcycle accidents. Most existing approaches follow a two-stage pipeline in which motorcycles and riders are first detected, followed by classification of helmet usage. Recent advancements in object detection and image classification have enabled reliable helmet detection even in crowded traffic scenes [1], [2].

Another important traffic violation is triple riding, which poses a serious safety risk for motorcycle users. Deshpande et al. developed a deep-learning-based framework capable of identifying motorcycles, counting riders, and detecting triple-riding violations from surveillance footage [3]. Their work demonstrated that rider-count estimation can be effectively integrated with object detection systems, although challenges such as occlusion,

varying camera angles, and image quality continue to affect performance.

The detection of mobile phone usage while riding or driving has also become an active area of research. Distracted driving caused by mobile phone usage contributes significantly to road accidents. Charef et al. proposed a computer vision approach based on YOLOv8 for detecting mobile phone usage among motorcycle riders [4]. Their findings highlighted the difficulty of accurately identifying handheld devices due to object size, rider movement, and partial occlusions. Nevertheless, recent advances in deep learning have shown promising results for real-time distraction detection systems.

Automatic Number Plate Recognition (ANPR) is a critical component of intelligent traffic management systems. ANPR systems automatically detect vehicle registration plates and extract textual information using image processing, optical character recognition (OCR), and deep learning techniques. Modern ANPR solutions commonly employ object detection models such as YOLO for plate localization and convolutional neural networks for character recognition, achieving high accuracy under varying environmental conditions [5].

Several countries have integrated ANPR systems with electronic traffic enforcement mechanisms to automate violation management. E-Challan systems digitally generate traffic tickets based on detected violations and maintain centralized violation records. Such systems reduce manual intervention, improve transparency, and increase enforcement efficiency. Automated enforcement frameworks combining violation detection, ANPR, and digital ticket generation have been successfully deployed in various smart traffic management applications [6].

Despite significant progress in individual components such as helmet detection, triple-riding detection, mobile phone detection, and ANPR, most existing studies focus on a single violation type. Comprehensive systems capable of detecting multiple violations and

automatically generating digital challans remain relatively limited. Furthermore, the performance of these systems often depends on the availability of high-quality datasets that accurately represent real-world traffic conditions. Therefore, the development of an integrated multi-violation detection and automated e-challan system remains an important research area in intelligent transportation systems.

3 Proposed Methodology

a. Overview

The proposed Automated Two-Wheeler Violation Detection and E-Challan System is designed to automatically monitor traffic activities and identify two-wheeler traffic violations in real time using computer vision and deep learning techniques. The system processes live CCTV or RTSP video streams, detects motorcycles and riders, identifies traffic violations such as helmet non-compliance and triple riding, extracts vehicle registration numbers, and generates violation records for further enforcement actions.

The methodology follows a modular pipeline architecture consisting of video acquisition, object detection, violation analysis, number plate recognition, violation evidence generation, and data management. This modular approach improves scalability, maintainability, and real-time performance

b. Software Development Life Cycle (SDLC)

Our project utilizes the Iterative Waterfall Model for its structured approach combined with the flexibility to revisit and refine previous phases based on testing feedback. This is essential for integrating deep learning models where data collection, training, and deployment require constant evaluation and tuning.

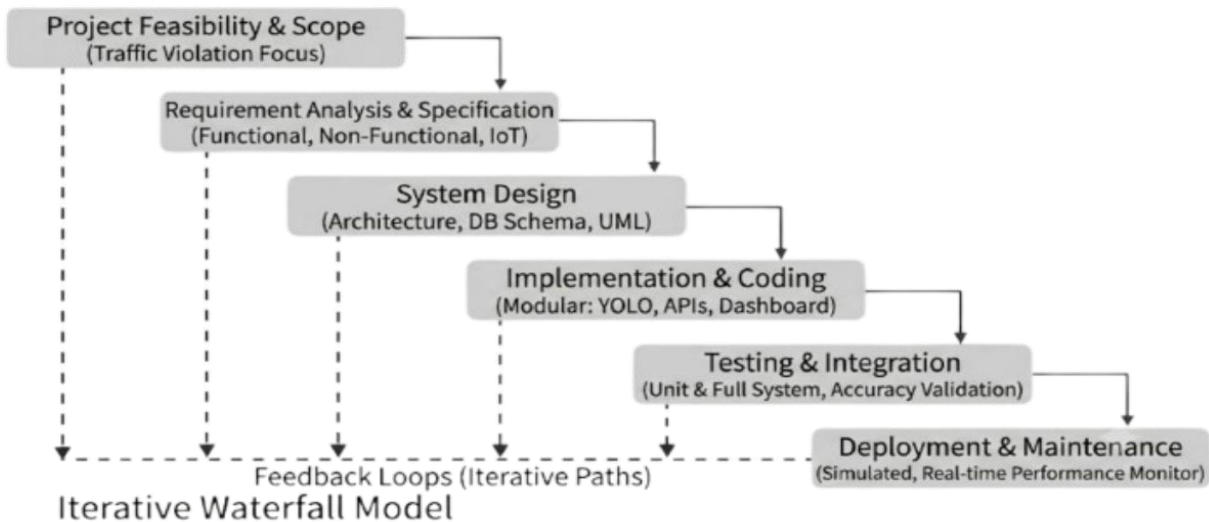


Figure 1: SDLC

Phases of the SDLC Model

- **Requirement Analysis & Specification:** In this initial phase, all functional and non-functional requirements are gathered. This includes defining the project scope, identifying targeted traffic violations (helmet non-compliance, triple pillion riding), and determining the technical specifications for edge cameras, server computation capacities, and user roles for Traffic Police.
- **System Design:** The architectural layout of the system is established here. It maps out the three-layer IoT architecture (Perception, Network, Application) and details the software components, database schemas (relational mapping), and UML diagrams such as the Use Case and Sequence Diagrams.
- **Implementation & Coding:** The system is developed in modular components during this phase. The computer vision pipeline (YOLO and tracking algorithms) is trained and configured, backend APIs are built using Node.js/Express.js, the PostgreSQL database is structured, and the responsive admin dashboard interface is engineered.
- **Testing & Integration:** Individual modules are first unit-tested, followed by full system integration testing. The AI Engine's detection accuracy (Precision/Recall) is evaluated against validation video clips, and the communication loop between the Edge, Central Server, Police Dashboard, and Notification API is thoroughly checked for latency and bugs.
- **Deployment & Maintenance:** The final refined system is deployed in a simulated environment. The backend and dashboard are hosted, the database is launched, and the camera feed ingestion pipeline is finalized. Ongoing maintenance involves monitoring model performance under different lighting conditions and updating core registries.

c. Dataset Acquisition

The effectiveness of deep learning models is highly dependent on the quality, diversity, and relevance of the training datasets used during model development. For this project, publicly available datasets related to helmet detection, motorcycle and rider detection, triple-riding detection, mobile phone usage detection, and automatic number plate recognition (ANPR) are utilized. Using existing datasets enables efficient model training, validation, and evaluation without the need for extensive field data collection.

The selected datasets contain images and annotated samples representing various traffic scenarios, including riders wearing and not wearing helmets, motorcycles carrying multiple passengers, riders using mobile phones while riding, and vehicle number plates captured under

different environmental conditions. These datasets provide the necessary data required to train the individual modules of the proposed system.

The datasets used in this study are listed in Table 3.1.

Table 3.1: Publicly Available Datasets Used in the Proposed System

Dataset Name	Purpose	Source/Link
Helmet Detection Project	Helmet Detection	https://universe.roboflow.com/nckh-2023/helmet-detection-project
Motorcycle-Rider Dataset	Motorcycle and Rider Detection	https://universe.roboflow.com/srishti-singh/motorcycle-rider-o5jx6
Triple Ride Detection Dataset	Triple Riding Detection	https://universe.roboflow.com/harsh-as-workspace/triple-ride-detection
License Plate Recognition Dataset	Number Plate Detection	https://universe.roboflow.com/roboflow-universe-projects/license-plate-recognition-rxg4e
License Plate Recognition (OCR) Dataset	OCR Training and Validation	https://www.kaggle.com/datasets/adilshamim8/license-plate-recognition
Rider Using Mobile Detection Dataset	Mobile Phone Usage Detection	https://universe.roboflow.com/vishnu-hje03/phone-zz2qe

The Roboflow datasets provide annotated images that can be directly exported in YOLOv8-compatible formats, facilitating seamless integration with the object detection framework used in this project. The Kaggle dataset is utilized to improve the performance of the Optical Character Recognition (OCR) module responsible for extracting textual information from vehicle number plates.

d. Dataset Preparation and Annotation

Most publicly available datasets already contain annotations suitable for object detection and classification tasks. However, the datasets may require preprocessing and standardization before model training.

The preparation process includes:

- Dataset inspection and cleaning
- Removal of duplicate or corrupted samples
- Conversion of annotations into YOLO format
- Image resizing and normalization
- Dataset merging (if multiple datasets are used)
- Class label standardization

Where necessary, additional annotations may be created using annotation tools such as Roboflow or LabelImg to ensure consistency across datasets.

The prepared dataset will be divided into three subsets:

- Training Set (70%)
- Validation Set (20%)
- Testing Set (10%)

To improve model generalization, data augmentation techniques such as horizontal flipping, rotation, brightness adjustment, scaling, and cropping may be applied during training.

e. Vehicle and Rider Detection

The first stage of the system involves detecting motorcycles and riders from video frames. A YOLOv8 object detection model is utilized due to its high accuracy and real-time processing capability. The model processes each video frame and generates bounding boxes around

motorcycles and riders.

The detection process includes:

- Frame Extraction
- Image Preprocessing
- Object Detection
- Bounding Box Generation
- Confidence Filtering
- The detected motorcycles are forwarded to subsequent modules for violation analysis.

f. Helmet Detection

Helmet detection is performed after rider localization.

The detected rider region is cropped and passed through a helmet classification model. The model determines whether the rider is wearing a helmet.

Possible outputs include:

- Helmet Detected
- No Helmet Detected

If a rider is found without a helmet, the system flags the event as a violation and forwards it to the violation processing module.

g. Triple Riding Detection

Triple riding detection is implemented by analyzing the number of riders associated with a motorcycle.

After motorcycle detection:

- Rider instances are identified.
- Riders are associated with the detected motorcycle.
- The total rider count is calculated.

Decision rules:

- 1 Rider $\rightarrow$ Normal

- Riders → Normal
- More than 2 Riders → Triple Riding Violation

When the rider count exceeds the permitted limit, a violation is generated.

h. Number Plate Detection and Recognition

Vehicle identification is performed through Automatic Number Plate Recognition (ANPR).

The process consists of two stages:

Number Plate Detection

YOLOv8 is used to localize the vehicle registration plate.

Character Recognition

The detected plate region is cropped and passed to EasyOCR or PaddleOCR for text extraction.

The extracted text is then cleaned and validated before being stored in the database.

The output includes:

- Vehicle Registration Number
- Confidence Score
- Timestamp

i. Violation Processing Engine

The Violation Processing Engine acts as the decision-making component of the system.

It receives information from:

Helmet Detection Module

Triple Riding Detection Module

ANPR Module

The engine evaluates predefined rules and determines whether a violation has occurred.

Example rules:

IF Helmet = False

THEN Violation = No Helmet

IF Rider_Count > 2

THEN Violation = Triple Riding

When a violation is confirmed, evidence is generated automatically.

j. Evidence Generation

For each detected violation, the system generates evidence containing:

- Violation Type
- Vehicle Number
- Captured Image
- Date
- Time
- Camera Location
- Confidence Score

The evidence is stored for future verification and enforcement purposes.

k. Database Design

All violation records are stored in a centralized database.

The database maintains:

Vehicle Information

- Vehicle Number
- Detection Time
- Camera ID

Violation Information

- Violation Type
- Evidence Image
- Confidence Score

System Logs

- Detection Events
- User Actions
- Alert History

The database enables efficient retrieval and analysis of violation records.

I. Dashboard and Monitoring System

A web-based dashboard is developed using FastAPI and modern frontend technologies.

The dashboard provides:

- Live Video Monitoring
- Violation Alerts
- Search by Vehicle Number
- Violation History
- Analytics and Reports

Traffic authorities can review detected violations and monitor system performance through the dashboard.

m. E-Challan Generation Module

Once a violation is verified, the system automatically generates an electronic challan.

The e-challan contains:

- Challan Number
- Vehicle Number
- Violation Type
- Date and Time
- Evidence Image
- Fine Amount

The generated challan can be stored digitally and forwarded to the appropriate enforcement authority.

n. System Workflow

The overall sequential logic and operational behavior of the proposed system are integrated into a unified pipeline. The system processes live video feeds in real-time, extracts features, evaluates compliance rules, and executes administrative actions. The step-by-step workflow of this automated pipeline is summarized below:

- **Video Acquisition:** Capture the live video stream from the traffic surveillance camera.
- **Frame Extraction:** Segregate the continuous video stream into individual frames for processing.
- **Object Detection (Vehicles):** Scan frames to detect motorcycles and isolate the riders.
- **Safety Compliance Check (Helmet):** Analyze the isolated riders to detect helmet usage.
- **Load Compliance Check (Triple Riding):** Count the number of concurrent riders on a single motorcycle to detect triple-riding violations.
- **License Plate Localization:** Detect and bounding-box the vehicle's number plate if a violation (no helmet or triple riding) is suspected.
- **Character Recognition:** Perform Optical Character Recognition (OCR) on the localized number plate to extract the registration text.
- **Rule Evaluation:** Process the combined data through the system's decision logic to finalize the violation status.
- **Evidence Generation:** Crop violation frames and log timestamps to compile immutable evidence.
- **Data Persistence:** Store the violation information, vehicle details, and evidence paths securely in the central database.
- **Action Execution:** Automatically generate an e-challan linked to the vehicle owner's registration.
- **User Interface Update:** Display the real-time violation alerts and metrics on the system dashboard.

To map out the concurrent processing, decision loops, and data-saving thresholds of this pipeline, the complete behavioral flow is illustrated in the activity diagram below (see Figure 2).

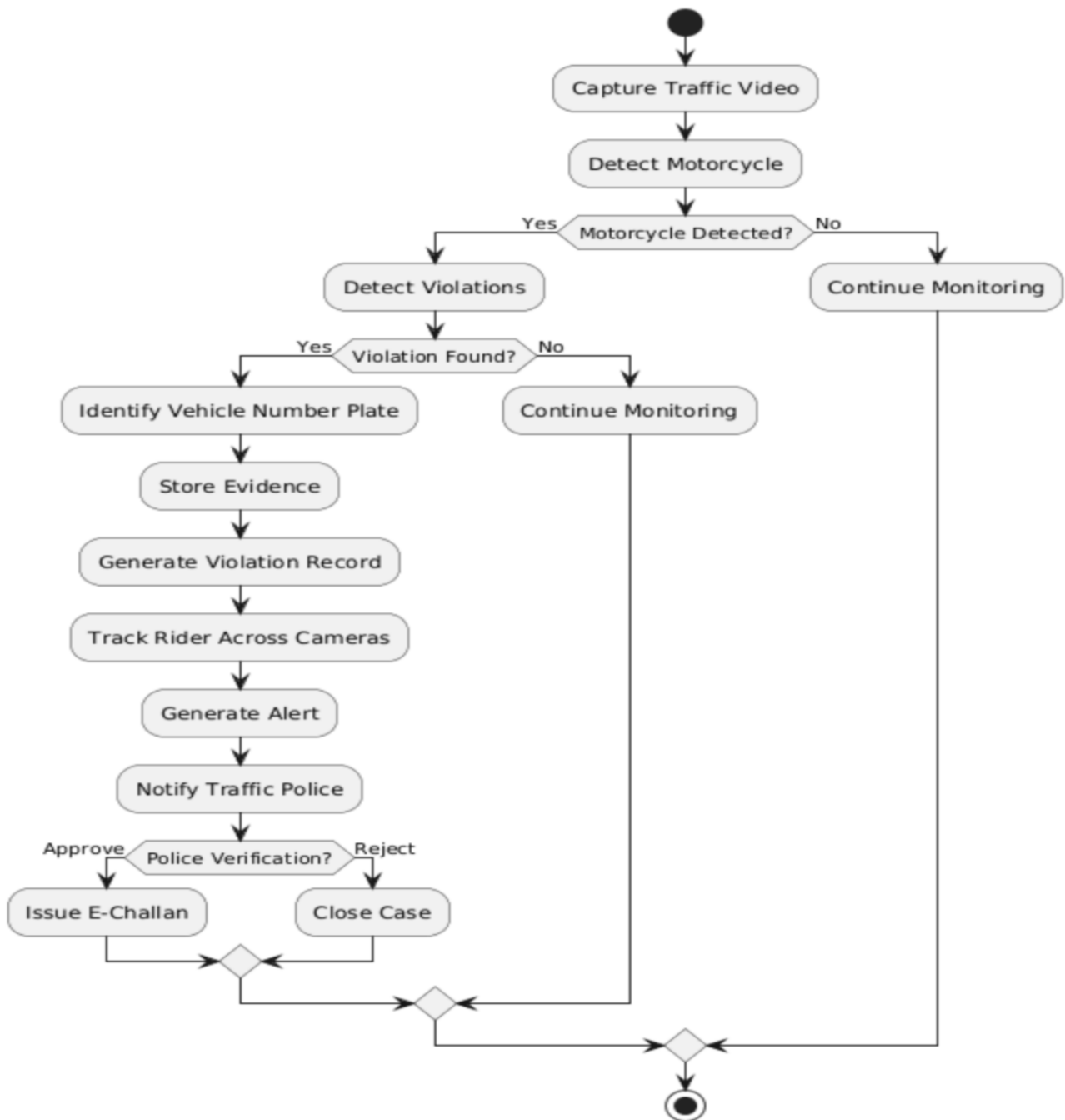


Figure 2: System Workflow

As shown in Figure 2, the system operates as a continuous loop. Frame extraction and object detection happen sequentially, whereas helmet verification and rider counting can execute in parallel paths. The decision diamond acts as a gatekeeper: if no violation rules are breached, the system discards the current frame buffer and immediately loops back to process the next incoming frame, optimizing processing overhead.

4 System Design

a. Functional Requirements

- **Real-time Video Ingestion:**

The system must continuously stream and ingest raw video data from distributed intersection cameras.

- **Automated Violation Screening:**

The machine learning models must track vehicle trajectories and monitor lanes in real time.

The system must automatically flag driving anomalies such as speeding or red-light infractions.

- **Targeted License Plate Extraction (ANPR):**

The system must trigger license plate recognition and character extraction only after an infraction is validated.

The system should isolate the vehicle bounding box and capture a high-definition image snapshot for evidence.

- **Challan Suggestion and Verification:**

The system must notify the traffic police dashboard immediately after a violation is flagged.

The system must match the plate characters against the registration database and automatically generate a suggested Challan with calculated fine amounts.

The platform must provide an administrative interface for traffic personnel to review the video evidence, plate text, and either approve or reject the suggested Challan.

- **Automated Notifications:**

The platform must send instant notifications (SMS/Email) containing a summary of the violation and a secure payment link to the registered vehicle owner.

b. Non-Functional Requirements

- **Performance & Latency:**

The localized computer vision models must process individual video frames with an inference latency of under 30ms.

- **System Accuracy:**

The Automatic Number Plate Recognition (ANPR) subsystem must achieve a minimum character accuracy rating of 95% across daytime, nighttime, and adverse weather conditions.

- **Scalability:**

The cloud application backend must support simultaneous incoming alert payloads from up to 500 edge camera units without service degradation.

- **Data Security & Privacy:**

All captured evidence media, including vehicle images and raw video segments, must be encrypted using AES-256 protocols to prevent unauthorized tampering.

- **High Availability:**

The central application server and reporting databases must guarantee a system uptime profile of 99.9%, excluding scheduled off-peak maintenance blocks.

5 System Design Architecture

The proposed system consists of the following major components:

- Video Acquisition Module
- Vehicle and Rider Detection Module
- Helmet Detection Module
- Triple Riding Detection Module
- Mobile Usage Detection Module
- Number Plate Detection and Recognition Module
- Violation Processing Engine
- Database and Evidence Storage Module
- Monitoring Dashboard and Reporting Module

The video stream captured from CCTV cameras is continuously processed by the AI engine. Detected objects are analyzed to determine whether any traffic violation has occurred.

When a violation is detected, relevant information including images, timestamps, and vehicle details is stored in the database and displayed through the monitoring dashboard.

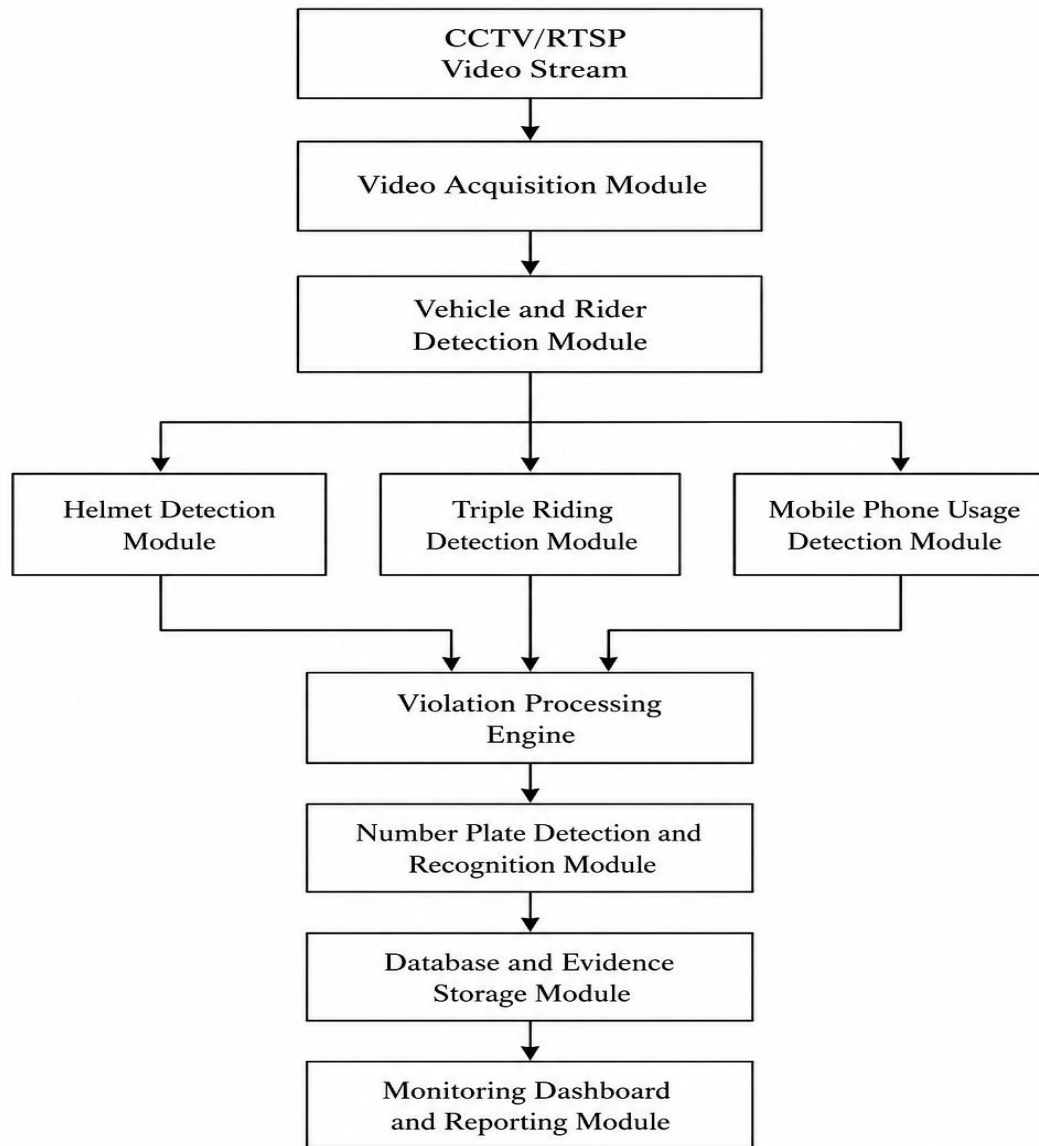


Figure 3: System Architecture

a. Use Case Diagram

This Use Case Diagram illustrates the functional boundaries of the automated traffic system by mapping interactions between external actors and core system operations. It shows the sequential

edge automation flow initiated by the camera—including video capture, violation detection, and vehicle identification—alongside administrative workflows executed by the System Admin and Traffic Police, such as evidence verification, E-Challan issuance, and automated alerting.

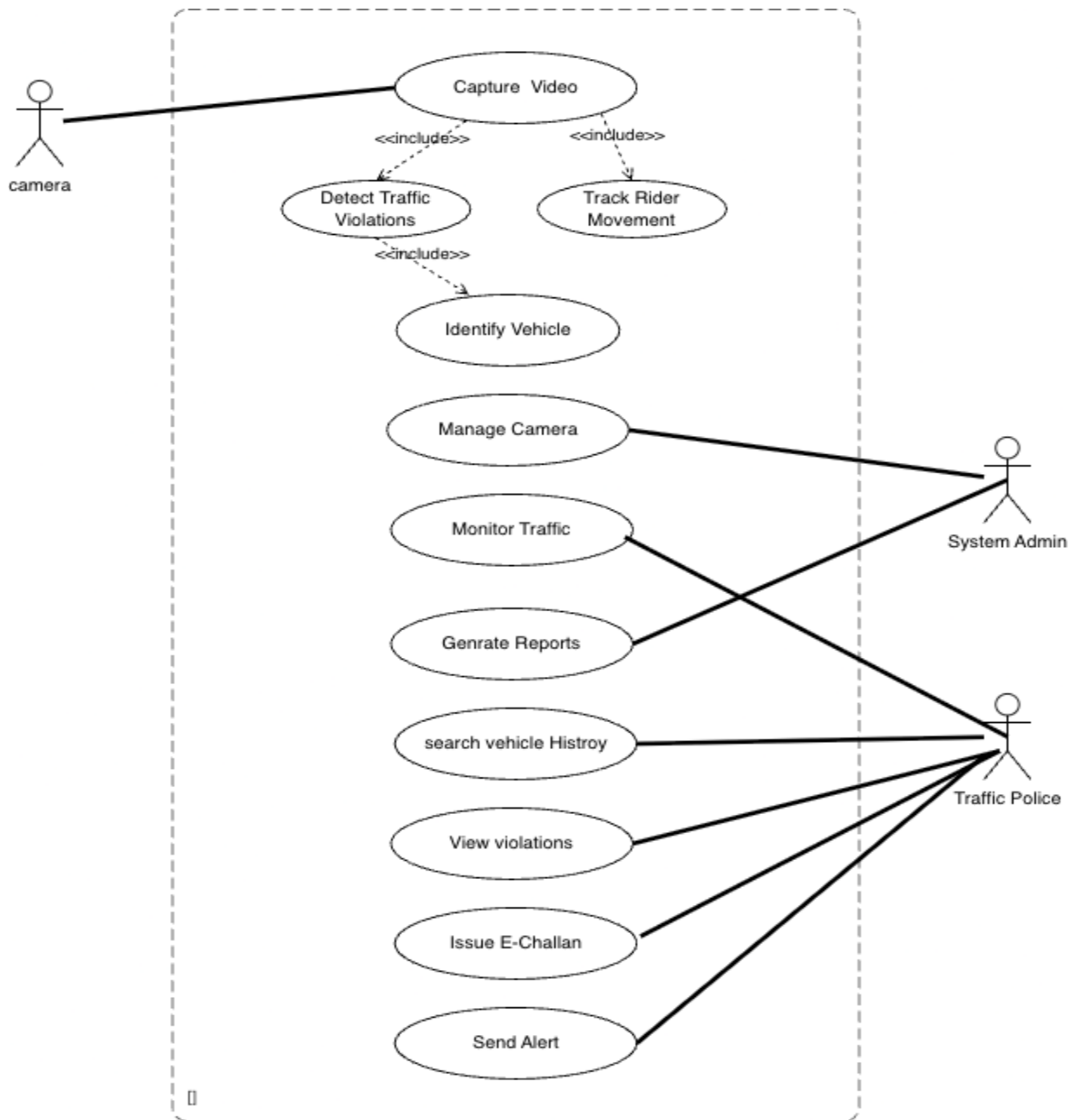


Figure 4: Use Case Diagram

b. Entity Relationship Diagram (ERD)

This Entity Relationship Diagram (ERD) defines the relational database schema for the traffic control system, mapping ownership and infrastructure records to incident workflows. It links

owners and vehicles through a historical mapping table while structuring how traffic violations tie back to specific cameras, fine categories, evidence media, and final E-Challan invoices.

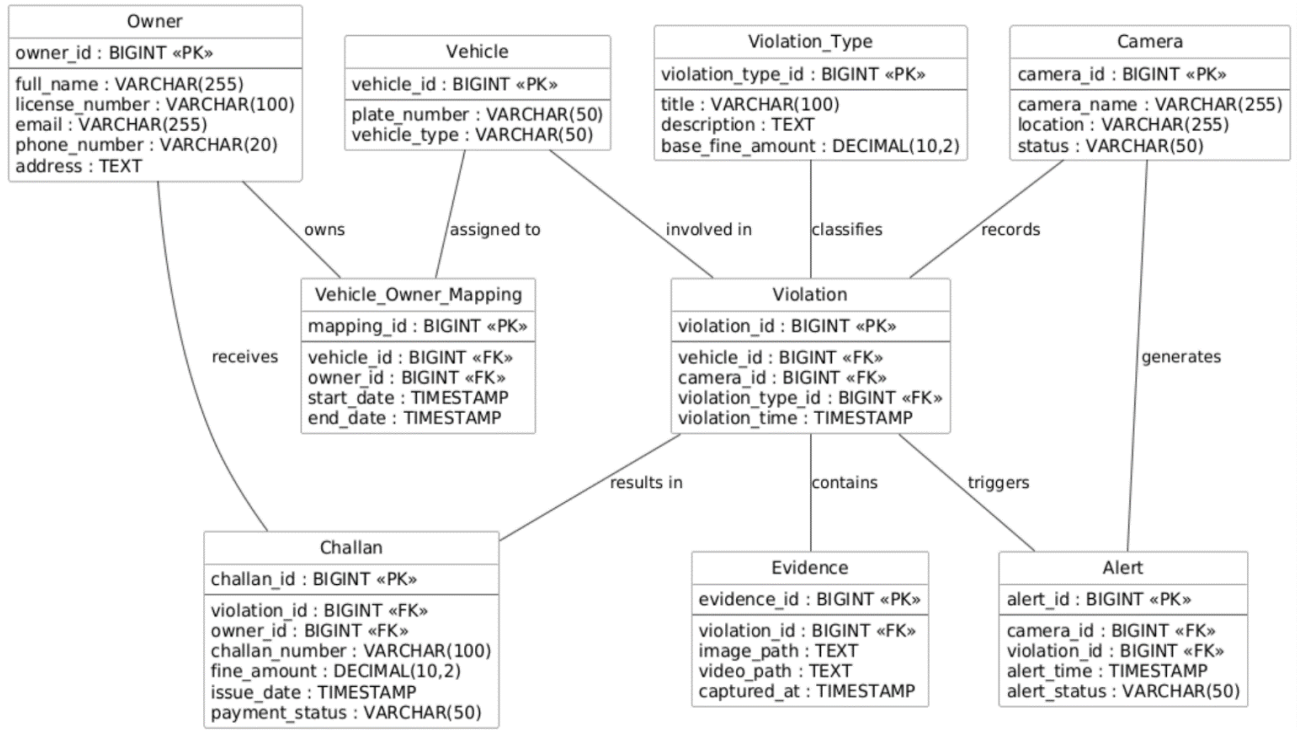


Figure 5: ER Diagram

C. Class Diagram

This Class Diagram defines the object-oriented structure of the system, mapping core domain entities and their structural attributes into clean backend data models. It represents how core

entities like owner Vehicle, and camera interact to produce a central violation object, which subsequently generates specific Evidence records, downstream messaging Alert blocks, and final challan billing instances.

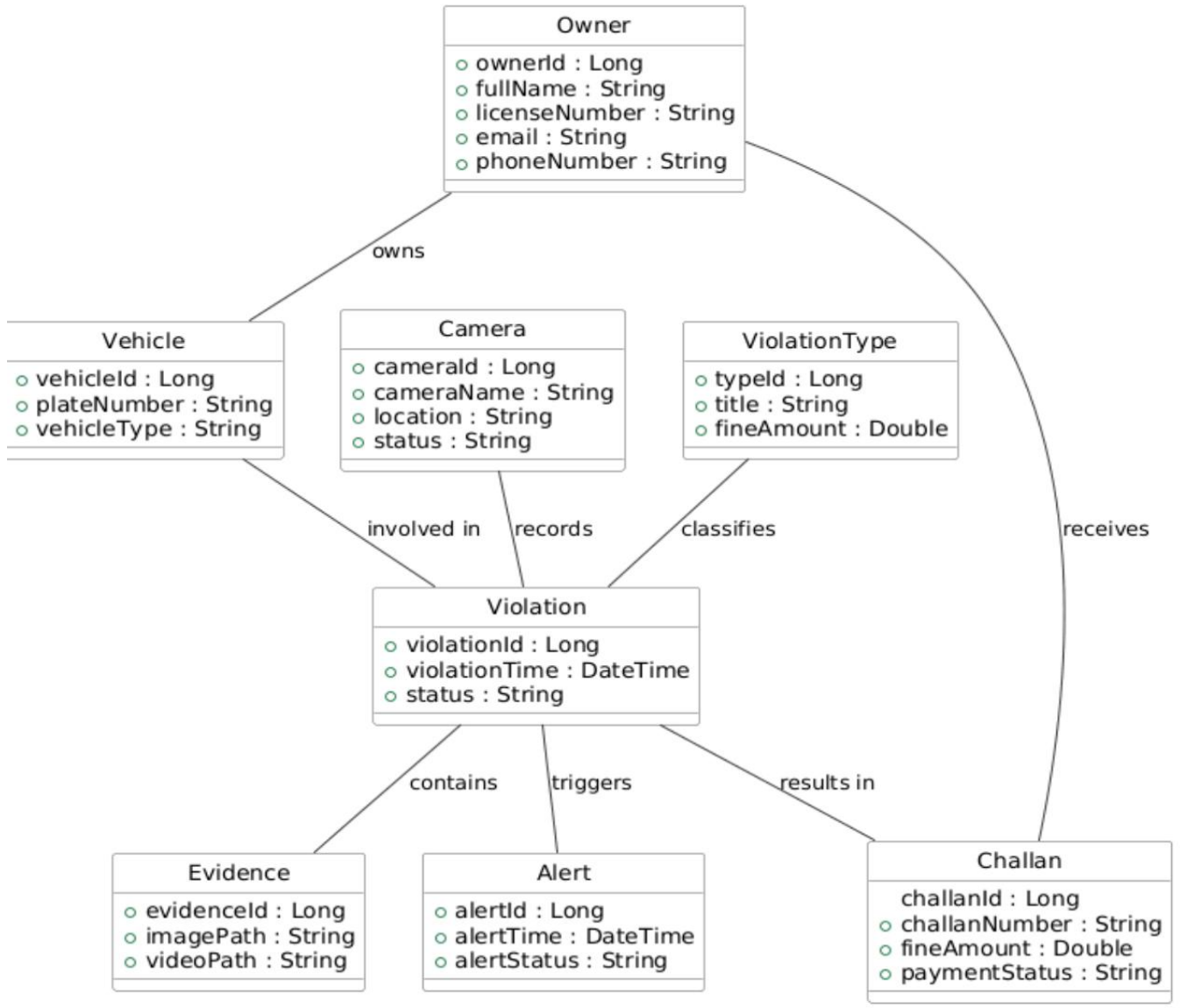


Figure 6: Class Diagram

d. Component Diagram

This Component Diagram illustrates the modular design and high-level architectural blocks of the system, showcasing how various software subsystems interact with human operators. It

models the modular progression from initial data ingestion at the Traffic Camera through the edge-based AI Detection System (comprising Violation Detection, OCR, and Rider Tracking components) down to the centralized Management System, which handles alerting, verification interfaces for Traffic Police, and data persistence.

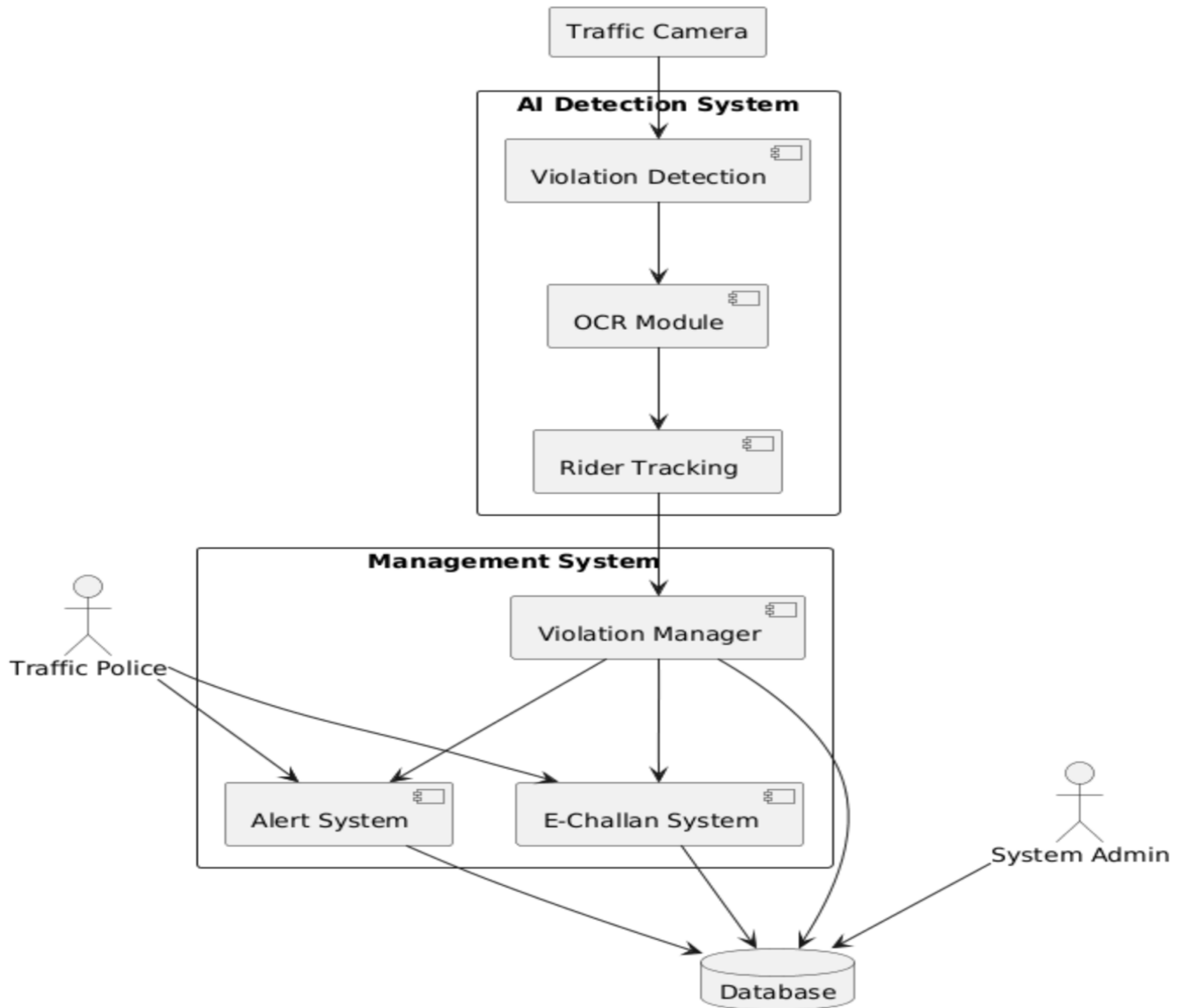


Figure 7: Component Diagram

e. Sequence Diagram

This Sequence Diagram outlines the step-by-step messaging timeline within the system core during a traffic infraction lifecycle across detection, processing, and final issuance phases. It models how the automated edge detection alerts are routed through the Central Server to a

human-in-the-loop validation step by Traffic Police before officially updating the database status and notifying the vehicle owner.

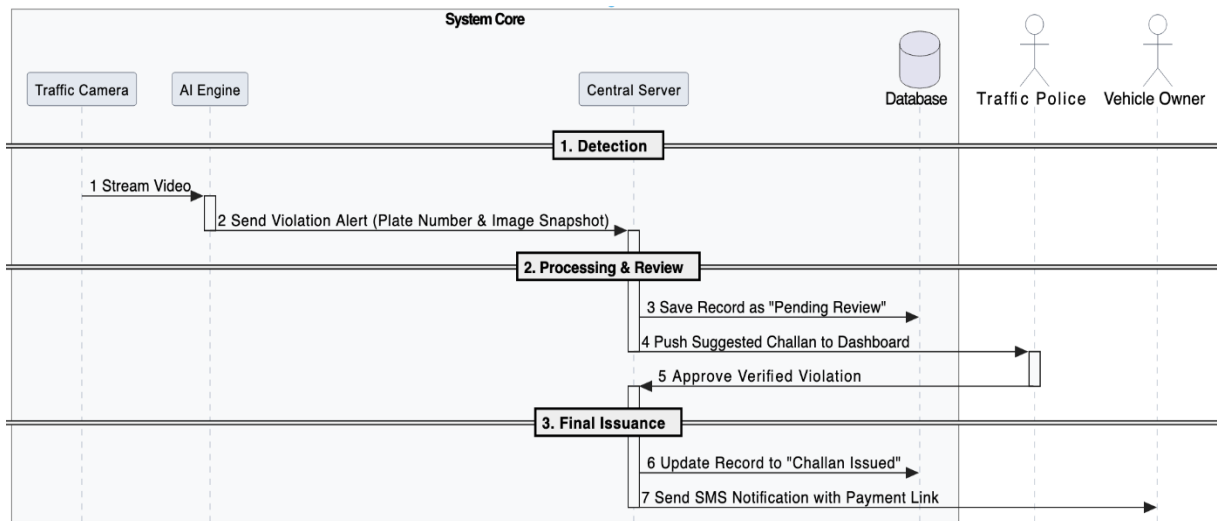


Figure 8: Sequence Diagram

6 Tools and Technologies to Be Used:

Frontend Development

- **React** – Used to build an interactive and responsive web dashboard for traffic administrators and police personnel to monitor incidents and verify violations.
- **Tailwind CSS** – For modern, utility-first styling of frontend dashboard components ensuring cross-device adaptability.

Backend Development

- **Node.js / Express.js** – Backend runtime and framework deployed on the central server to handle business logic, process inbound IoT payloads, route telemetry data, and manage API endpoints.

Database

- **SQL Database** – A robust relational database used to store structured data such as vehicle registries, owner identities, historical infraction entries, and official E-Challan records.

Machine Learning and Computer Vision

- **YOLO (You Only Look Once)** – Employed as the primary deep learning object detection framework on the edge. It is utilized for real-time localization of motorcycles and riders within video frames, and subsequently handles violation screening (such as detecting riders without helmets or carrying triple pillions) with high frame-rate performance.
- **Automatic Number Plate Recognition (ANPR)** – A dedicated optical character recognition and localization pipeline used to segment the vehicle's license plate area from the captured evidence frames and transcribe the alphanumeric characters into standard database strings.
- **ByteTrack / DeepSORT** – A multi-object tracking algorithm integrated alongside the detection model to persistently track rider movement and maintain individual vehicle identities across video frames, preventing duplicate violation counts for a single incident.
- **PyTorch / OpenCV** – PyTorch serves as the core deep learning library used to optimize and run model inference, while OpenCV handles the upstream real-time video stream ingestion, frame pre-processing, bounding-box overlays, and image cropping.

Notification Integration

- **Email Gateway** – Integrated for dispatching instantaneous violation alerts and automated digital fine notifications containing secure payment links directly to vehicle owners

7 Project Task and Time Schedule

The development of the Automated Two-Wheeler Violation Detection and E-Challan System will be completed in different phases over a planned schedule of 12 weeks. Each team member will contribute to specific tasks according to their area of expertise, ensuring efficient project completion within the allotted timeframe.

a. Project Task Schedule

S.N	Task / Activity	Duration	Week
1	Requirement Gathering & Problem Analysis	1 Week	Week 1
2	Dataset Collection, Inspection & Preparation	1 Week	Week 2
3	System Design & Architecture Planning	1 Week	Week 3
4	Database Design & API Planning	1 Week	Week 4
5	Vehicle & Rider Detection Model Development (YOLOv8)	1 Week	Week 5
6	Helmet Detection & Triple Riding Detection Model Development	1 Week	Week 6
7	Number Plate Detection & OCR Integration	1 Week	Week 7
8	Backend API Development & Database Integration	1 Week	Week 8
9	Frontend Dashboard Development	1 Week	Week 9
10	Violation Processing Engine & E-Challan Generation Module	1 Week	Week 10
11	System Integration, Testing & Debugging	1 Week	Week 11
12	Deployment, Final Report & Presentation Preparation	1 Week	Week 12

b. Work Distribution Among Group Members

Member	Responsibility
Member1	Frontend Development (Dashboard UI, Live Monitoring, Violation History, Analytics, Reports)
Member2	Backend Development (FastAPI, Database Design, APIs, Authentication, E-Challan Generation, System Integration)
Member3	Machine Learning Development (Dataset Preparation, YOLOv8 Training, Helmet Detection, Triple Riding Detection, ANPR & OCR Modules)

c. Gantt Chart



Figure 9: Gantt Chart

8 Expected Result and Outcomes

Expected Results

The system is expected to successfully ingest real-time video streams from roadside cameras and process them with minimal latency. The edge AI components should accurately identify motorcycles and detect targeted traffic infractions—such as helmet non-compliance or red-light violations—under varying ambient lighting and traffic densities. Upon detecting a violation, the system will reliably capture high-definition evidence frames, extract clear alphanumeric characters from license plates, and register a pending entry in the central database.

System Outcomes

- **Automated Violation Pipeline:** Complete reduction of manual traffic monitoring hours through automated edge-to-cloud incident dispatching.
- **Human-in-the-Loop Validation:** An intuitive, low-latency administrative dashboard that allows Traffic Police to verify evidence and approve or reject citations within seconds.
- **Efficient Fine Enforcement:** Automated generation and immediate delivery of digital E-Challans via Email, providing vehicle owners with an instantaneous, transparent payment link.
- **Reliable Data Persistence:** A fully normalized, secure relational database structure that safely logs historical ownership mappings, infrastructure health, and legal audit trails.

9 Conclusion

This project demonstrates the successful design and architecture of an intelligent, end-to-end Smart Traffic Control System. By combining edge-based computer vision engines with a robust relational database backend, the system bridges the gap between automated surveillance and legal enforcement.

The integration of an AI detection pipeline removes the need for continuous human oversight, while the human-in-the-loop verification step guarantees high citation accuracy and minimizes false positives. Ultimately, this system provides scalable infrastructure for modern urban environments, optimizing traffic law compliance, accelerating fine collection workflows, and fostering safer driving habits on public roads.

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